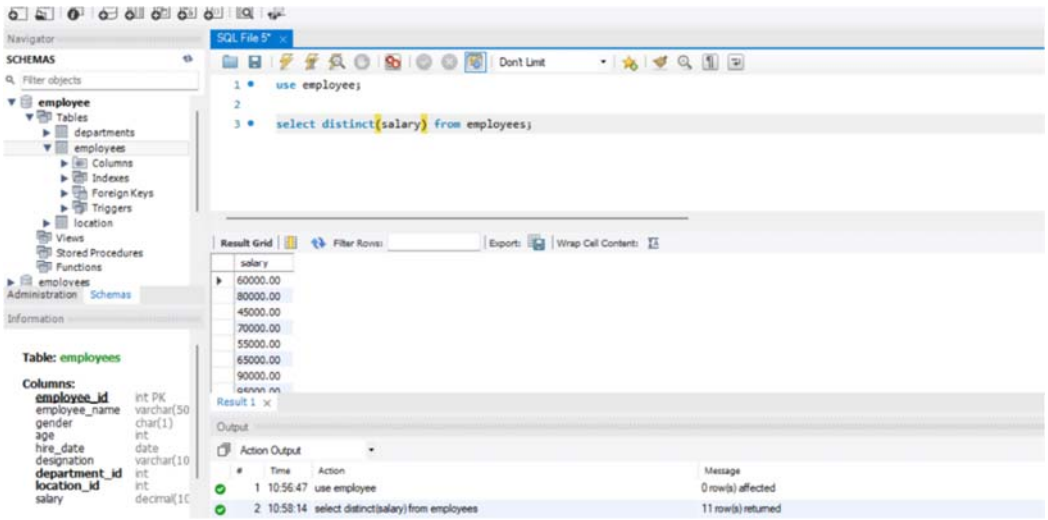
MySQL Assignment 2 – Querying Data

The employee data from the provided link was successfully inserted into the previously created departments, employees, and location tables using the given DML INSERT statements, and all table updates were completed successfully.

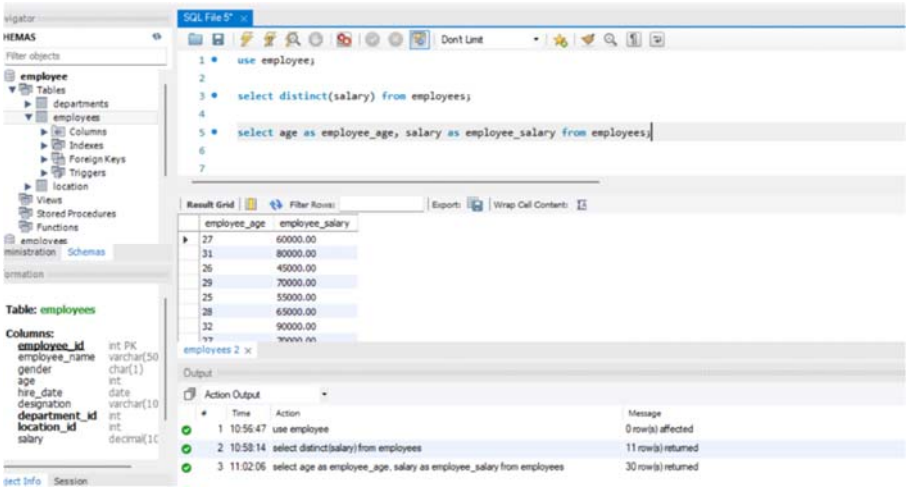
Distinct Values:

The distinct salaries from employees table were displayed using the query **SELECT distinct(salary) FROM employees;**



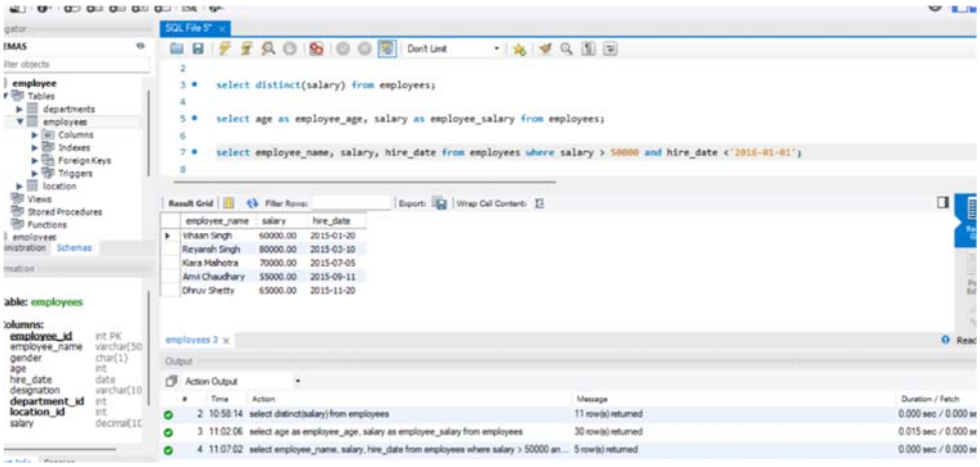
Alias (AS):

The Age and Salary columns were assigned the aliases Employee_Age and Employee_Salary, respectively, using the AS keyword.

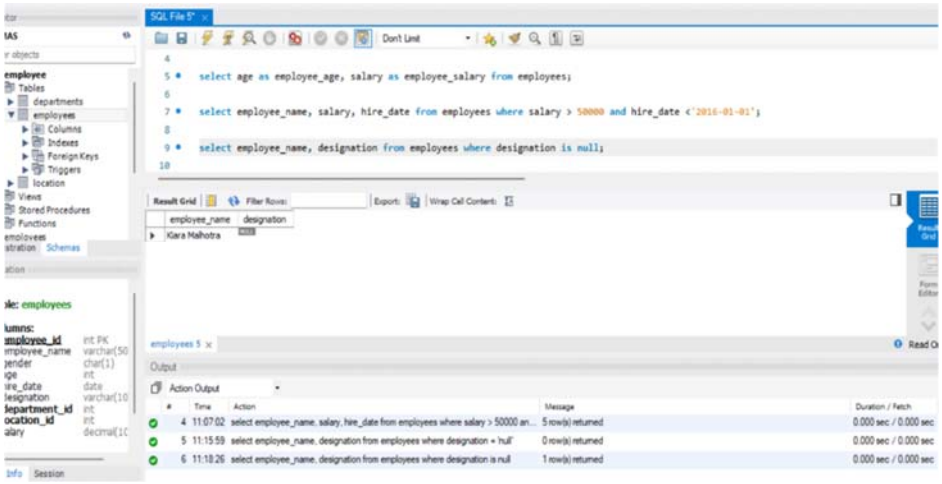


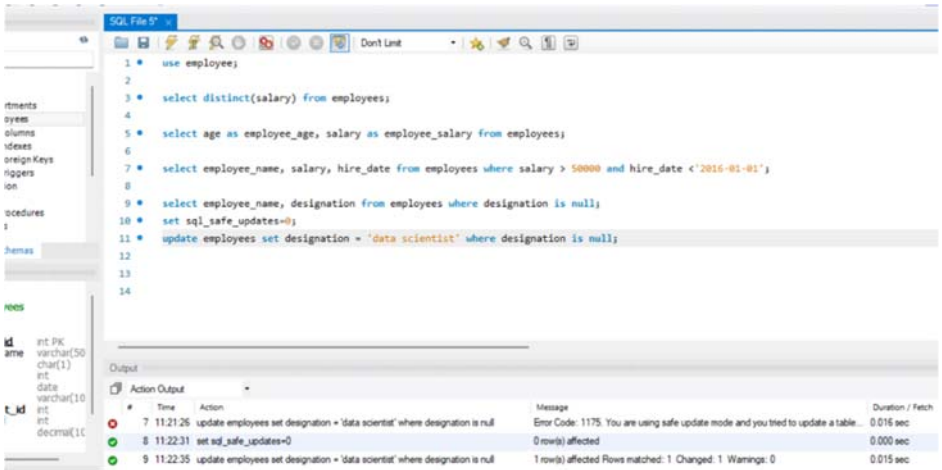
Where Clause & Operators:

Employee names, salaries, and hire dates were retrieved for employees whose salaries exceed 50,000 and whose hire dates are earlier than 2016-01-01, using the **WHERE** clause with the **AND** operator.



The employee record with a missing designation was identified using the **SELECT** query. The missing designation was then updated to 'Data Scientist' using the **UPDATE** statement with the **WHERE** clause. Safe update mode was temporarily disabled to allow the update operation.

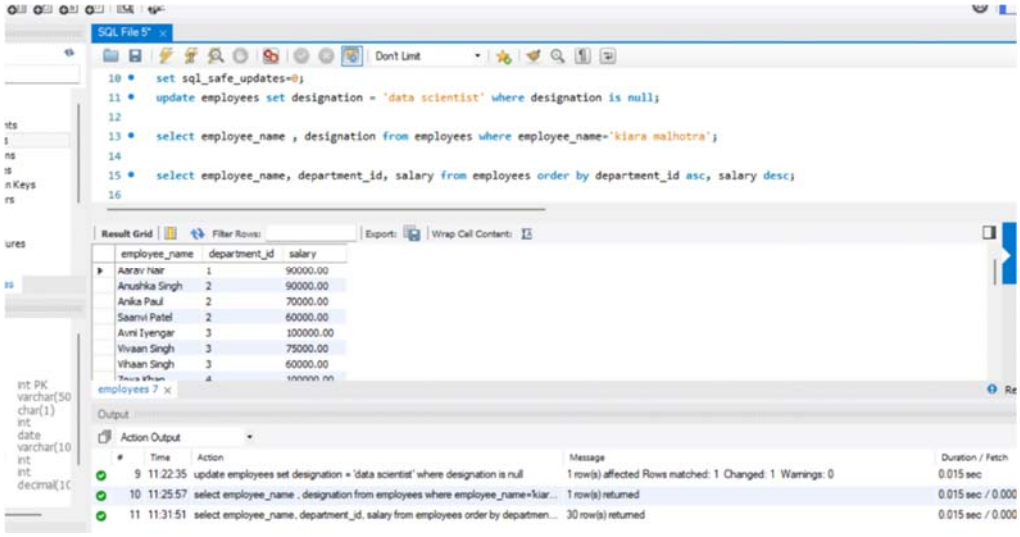




Sorting and Grouping Data:

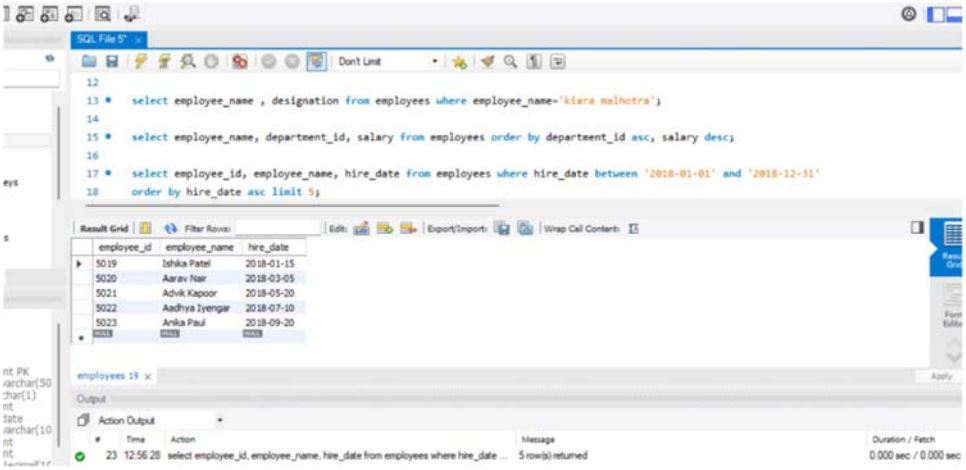
ORDER BY:

Employee records were sorted by Department_ID in ascending order and Salary in descending order using the ORDER BY clause.



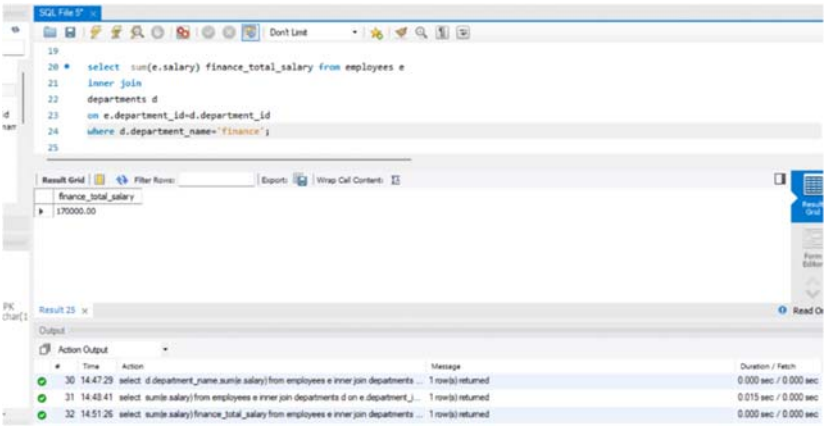
LIMIT:

The first five employees hired in the year 2018 were retrieved using the **WHERE** clause to filter employees by hire year and the **LIMIT** clause to display only five records.

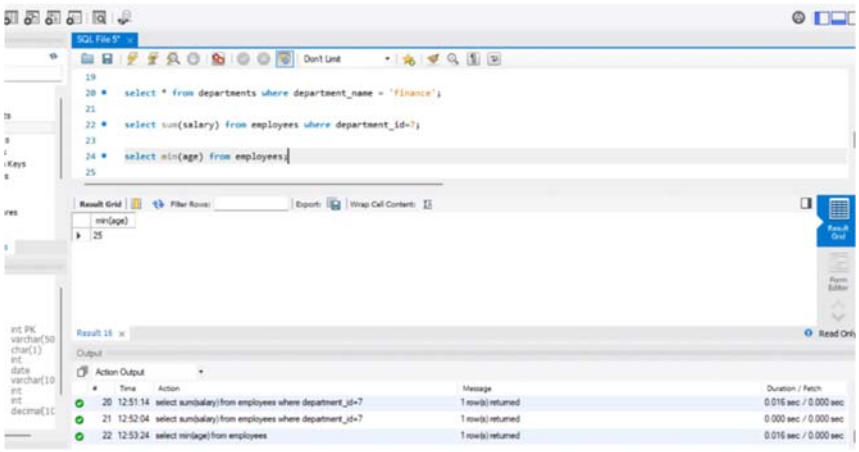


Aggregate Functions:

The total salary of employees working in the Finance department was calculated using the **SUM ( )** aggregate function by joining the Employees and Departments tables with an **INNER JOIN** and filtering records based on the department name.

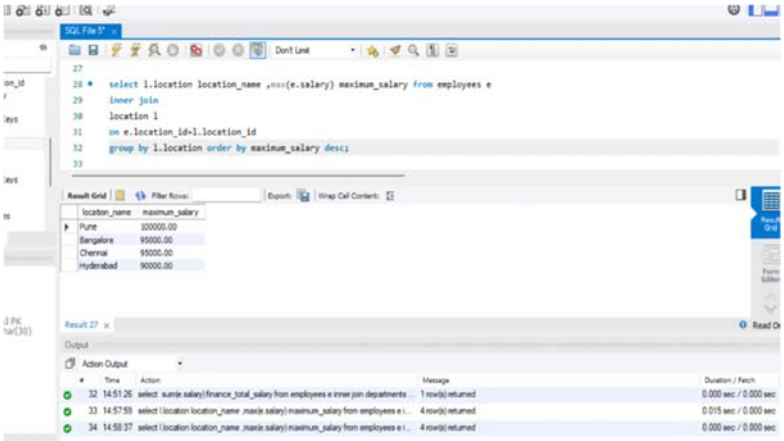


The minimum age among all employees was identified using the **MIN( )** aggregate function to determine the lowest value in the Age column.

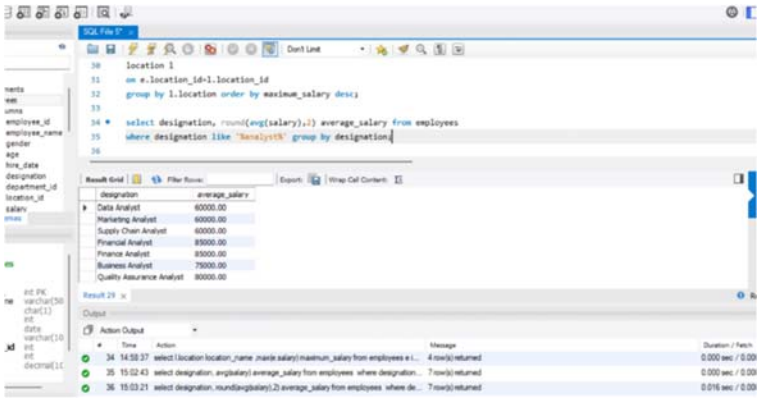


GROUP BY:

The maximum salary for each location was calculated using the **MAX( )** aggregate function, and the **GROUP BY** clause to group employees by location. The output displays each location along with its highest salary, sorted in descending order of maximum salary.

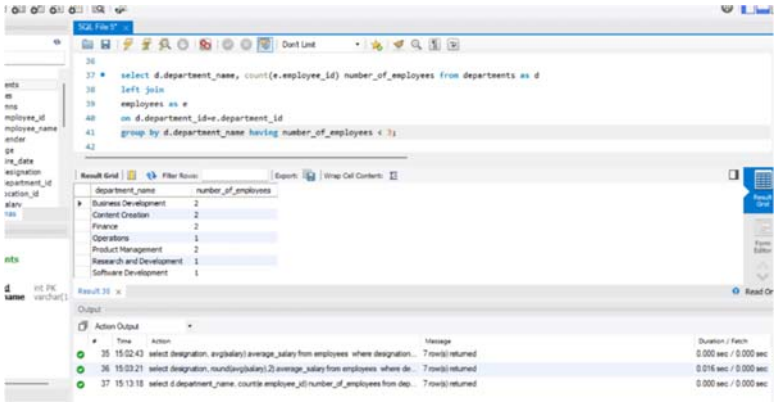


The average salary for each designation containing the word 'Analyst' was calculated using the **AVG()** aggregate function, with the **WHERE** clause to filter designations and the GROUP BY clause to group the results by designation.

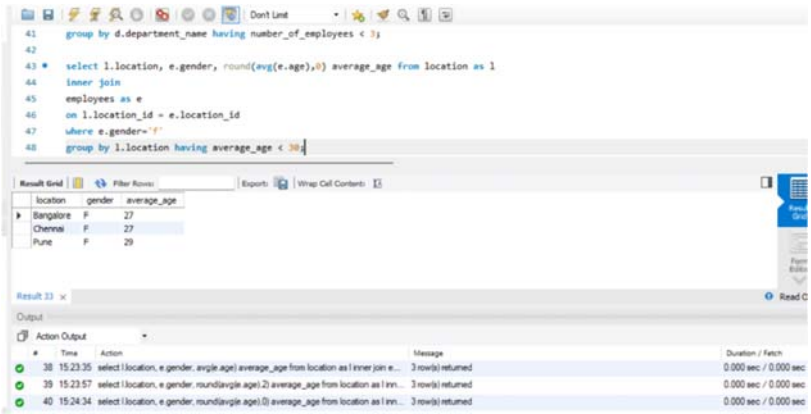


HAVING:

The department names and their corresponding employee counts were retrieved using a **LEFT JOIN**, **GROUP BY**, and **HAVING** clause to display only those departments with fewer than 3 employees.



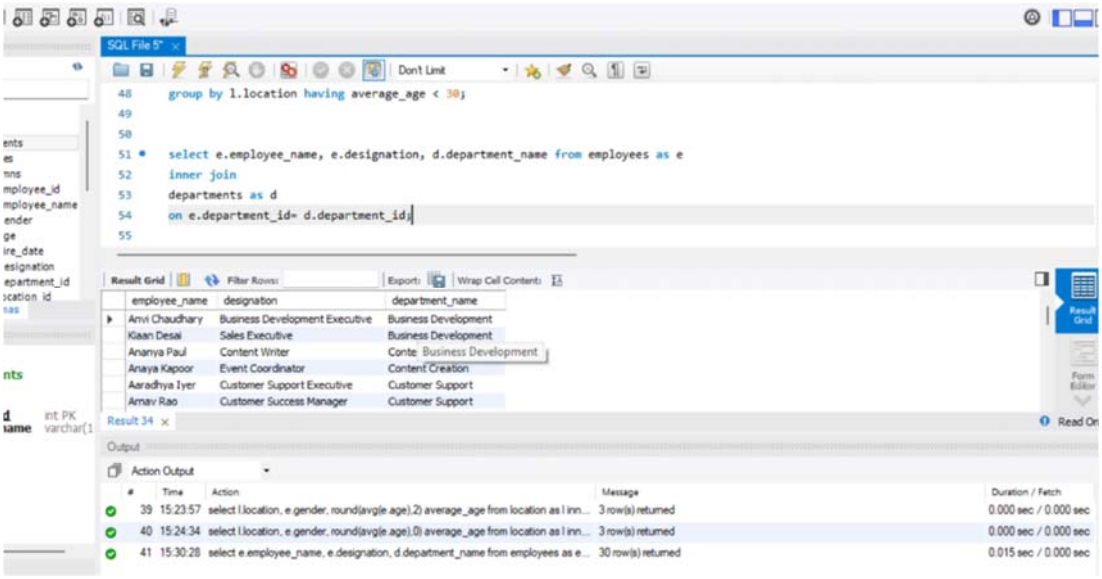
The average age of female employees was calculated for each location, and only those locations with an average age below 30 were displayed using the **INNER JOIN, WHERE, GROUP BY** and **HAVING** clauses.



Joins:

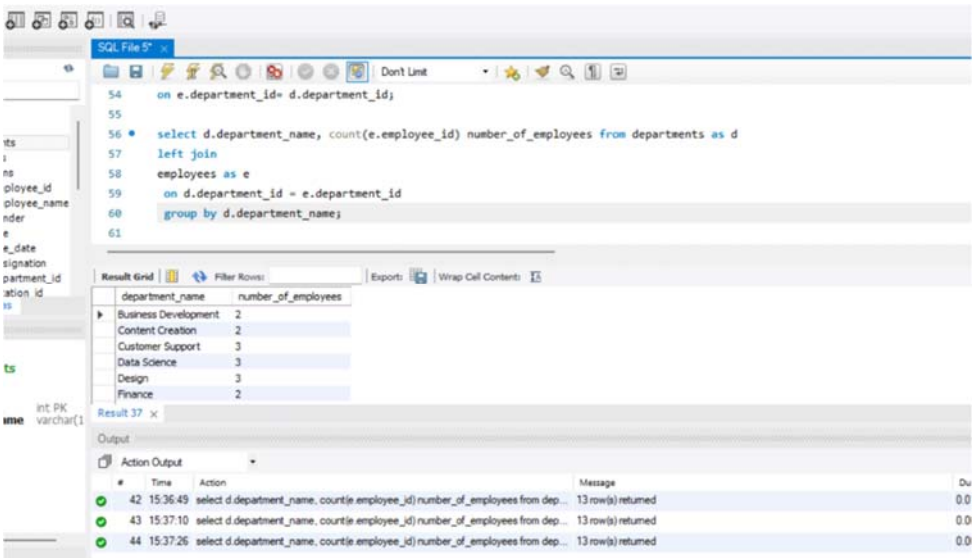
1. Inner Join:

The **INNER JOIN** clause was used to retrieve employee names, their designations, and corresponding department names by matching the department_id column in both the Employees and Departments tables. Only employees assigned to a department were displayed in the result.



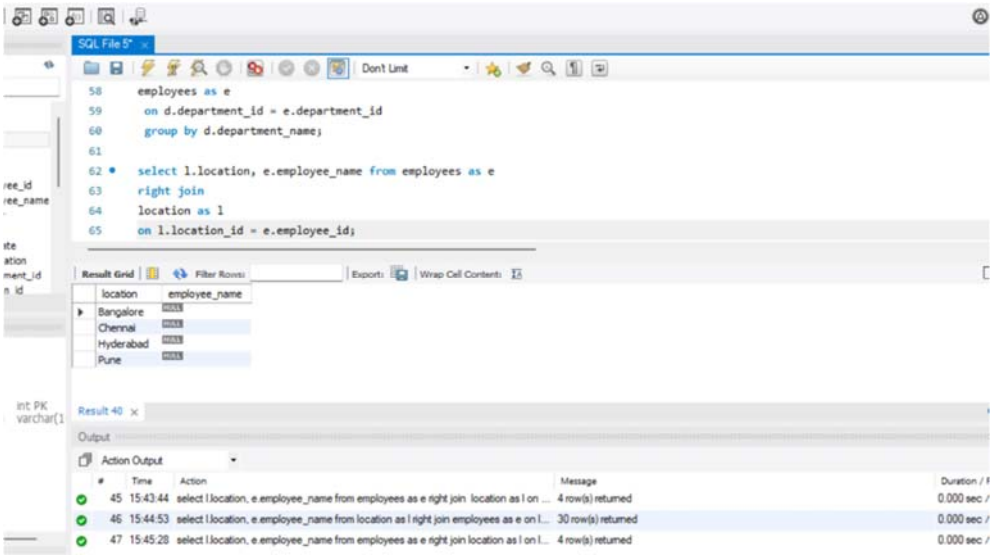
2. Left Join:

The **LEFT JOIN** clause was used to retrieve all department names along with the total number of employees in each department, including departments with no employees. The **COUNT()** aggregate function was used to calculate the number of employees in each department, and the **GROUP BY** clause was used to group the results by department name.



3. Right Join:

The **RIGHT JOIN** was used to retrieve all locations along with the names of employees assigned to each location. Locations without any assigned employees are also displayed, with **NULL** values for the employee names.



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